DIFFERENTIALS OF A MORPHISM LOCALLY OF FINITE PRESENTATION OF SPACES HAVE FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1264-1270 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1272-1277, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Upgrade of L16050 from finite type to finite presentation.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale-locality of the finite-presentation property for the module of differentials (lemma-localize-differentials) and the schemes case (Morphisms, lemma-finite-presentation-differentials).

Source reference. Original lemma 04CU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01V3: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1277–1277. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally of finite presentation, then $\Omega_{X/Y}$ is an $\mathcal{O}_X$-module of finite presentation.*

Proof. Follows from the schemes version, see Morphisms, Lemma 01V3 and étale localization, see Lemma 04CU. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.